GO BACN N

average probability of a bad channel -> π_0

average number of consecutive good slots $= \dfrac{1}{P_{01}} \Rightarrow P_{01} = \dfrac{1}{\text{AVG GOOD}}$

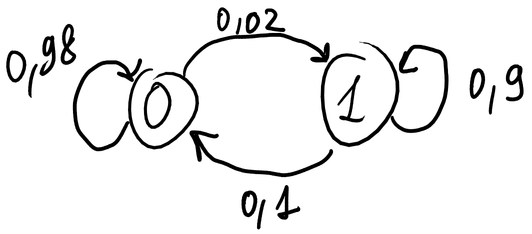
E4 Consider a two-state Markov channel with transition probabilities 0.98 (from the good state to itself) and 0.1 (from the bad state to the good state). The packet error probability is 1 for a bad slot and 0 for a good slot, respectively.

(a) Compute the throughput (average number of successes per slot) of a protocol that transmits packets directly on the channel, with no retransmissions.

(b) Compute the throughput of a Go-Back-N protocol if the round-trip time is $m = 2$ slots (i.e., a packet that is erroneous in slot t is retransmitted in slot $t+2$), in the presence of an error-free feedback channel.

(c) Same as in the previous point, with the difference that now the feedback channel is subject to iid errors with probability 0.1.

1) Calcolare la matrice di transizione per gli stati tramite i dati



$$C = \begin{pmatrix} 0,98 & 0,02 \\ 0,1 & 0,9 \end{pmatrix}$$

2)Calcolare i pi greco

$$\begin{cases} \pi_0 = 0,98\,\pi_0 + 0,1\,\pi_1 \\ \pi_1 = 0,02\,\pi_0 + 0,9\,\pi_1 \\ \pi_0 + \pi_1 = 1 \end{cases} \quad \begin{cases} \pi_0\,(1 - 0,98 + 0,1) = 0,1 \\ / \\ \pi_1 = 1 - \pi_0 \end{cases}$$

$$\begin{cases} 0,12\,\pi_0 = 0,1 & \Rightarrow \pi_0 = \dfrac{0,1}{0,12} = 0,833 \\ \pi_1 = 1 - \pi_0 & \Rightarrow \pi_1 = 0,167 \end{cases}$$

(a) Compute the throughput (average number of successes per slot) of a protocol that transmits packets directly on the channel, with no retransmissions.

Senza protocollo il troughput è uguale al pi0

$$\eta = \pi_0 = 0,833$$

(b) Compute the throughput of a Go-Back-N protocol if the round-trip time is $m = 2$ slots (i.e., a packet that is erroneous in slot t is retransmitted in slot $t+2$), in the presence of an error-free feedback channel.

1)Calcolare C^m, m è quasi sempre m = 2, quindi basta moltiplicare C per sé stessa

$$C^m = \begin{pmatrix} 0,98 & 0,02 \\ 0,1 & 0,9 \end{pmatrix} \times \begin{pmatrix} 0,98 & 0,02 \\ 0,1 & 0,9 \end{pmatrix} =$$

$$\begin{pmatrix} 0,98 \cdot 0,98 + 0,02 \cdot 0,1 & 0,98 \cdot 0,02 + 0,02 \cdot 0,9 \\ 0,1 \cdot 0,98 + 0,9 \cdot 0,1 & 0,1 \cdot 0,02 + 0,9 \cdot 0,9 \end{pmatrix}$$

$$C^m = \begin{pmatrix} 0{,}962 & 0{,}038 \\ 0{,}188 & 0{,}812 \end{pmatrix}$$

2) Adesso che abbiamo C^m possiamo usare la formula per il troughput in questo caso (o in alternativa calcolare il pi0 di nuovo, soltanto che dobbiamo usare la matrice aggiornata)

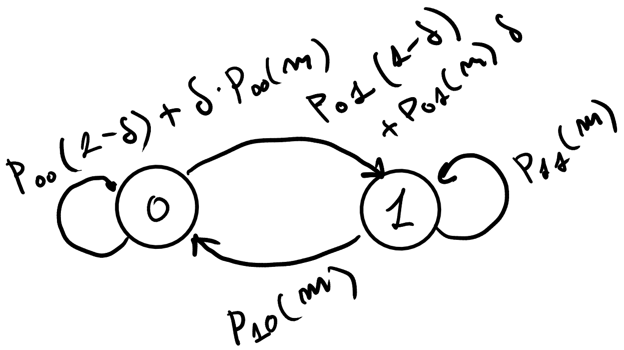
$$P = \begin{array}{c} \begin{pmatrix} 0{,}98 & 0{,}02 \\ 0{,}188 & 0{,}812 \end{pmatrix} \end{array} \begin{array}{l} \text{DA } C \\ \text{DA } C^m \end{array} \Rightarrow \begin{cases} \pi_0 = 0{,}98\,\pi_0 + 0{,}188\,\pi_1 \\ \pi_1 = 0{,}02\,\pi_0 + 0{,}812\,\pi_1 \\ \pi_0 + \pi_1 = 1 \end{cases}$$

OPPURE

$$\boxed{\eta = \frac{P_{10}(m)}{P_{10}(m) + m \cdot P_{01}}}$$

(c) Same as in the previous point, with the difference that now the feedback channel is subject to iid errors with probability 0.1. $\rightarrow \delta = 0{,}1$

In questo caso possiamo impostare la matrice con le nuove transizioni di stato:



$$\Rightarrow P = \begin{pmatrix} P_{00}(1-\delta) + \delta P_{00}(m) & \ldots \\ \ldots & \ldots \end{pmatrix}$$

e calcolarci di nuovo il pi0 col sistema, oppure:

$$\boxed{\eta = \frac{(1-\delta)\,P_{10}(m)}{(1-\delta+m\delta)\,P_{10}(m) + m\left[(1-\delta)P_{01} + P_{01}(m)\,\delta\right]}}$$

E4 Consider a two-state Markov channel with transition probabilities 0.98 (from the good state to itself) and 0.1 (from the bad state to the good state). The packet error probability is 1 for a bad slot and 0 for a good slot, respectively.

(a) Compute the throughput of a Go-Back-N protocol if the round-trip time is $m = 2$ slots (i.e., a packet that is erroneous in slot t is retransmitted in slot $t+2$), in the presence of an error-free feedback channel

(b) Consider now a system where a channel behavior as in point (a) (Markov model for the forward channel and error-free feedback channel) alternates with a channel behavior in which the forward channel is subject to iid errors with probability $\varepsilon = 0.01$. In particular, the channel follows the Markov model for a geometric number of slots with mean 1000000 slots, then follows the iid model for a geometric number of slots with mean 2000000 slots, then again the Markov model and so on. Compute the overall average throughput of the Go-Back-N protocol in this case.

$$\rightarrow \boxed{\eta = \frac{1-\varepsilon}{1-\varepsilon+m\varepsilon}}$$

TROUGHPUT IID FORWARD CHANNEL

$$\frac{1}{3} \cdot \eta_M + \frac{2}{3}\,\eta_{IID}$$

CASO ERRORI IID ANCHE IN FEEDBACK

$$\eta = \frac{(1-\delta)(1-\varepsilon)}{(1-\delta)(1-\varepsilon) + m\left[1-(1-\delta)(1-\varepsilon)\right]}$$

* RENEWAL CYCLE (1 SERVER)

E3 Consider a network node able to handle traffic at 10 Gbps under normal conditions. The node is subject to attacks, that arrive according to a Poisson process of rate $\lambda = 1/T_0$. For each attack, the node has a probability $1-\alpha$ of being infected, whereas with probability α the attack has no consequence. When a node gets infected, it automatically starts a clean-up process that lasts T_1 and occupies 70% of its resources, so that during this phase the node can only handle 3 Gbps. The clean-up process is successful with probability β (in which case the node starts working normally), whereas with probability $1-\beta$ it fails and the node needs to be restored manually by a human operator, which takes T_2, during which time the node does not handle any traffic. After being manually restored, the node starts working normally.

(a) By identifying an appropriate renewal cycle, compute the fraction of the time the node is not handling any traffic and the average traffic per unit time (in Gbps) handled by the node,

(b) Compute how often (e.g., how many times a day on average) a human operator's intervention is needed.

(For all the above quantities, find mathematical expressions as a function of the parameters, and then compute their numerical values for $T_0 = 20$ minutes, $T_1 = 20$ minutes, $T_2 = 3$ hours, $\alpha = 0.98$, $\beta = 0.9$.)

(a) By identifying an appropriate renewal cycle, compute the fraction of the time the node is not handling any traffic and the average traffic per unit time (in Gbps) handled by the node,

1)Dobbiamo calcolare il tempo di durata di un attacco

FREQUENZA ATTACCHI

$$\text{POISSON PROCESS CON } \lambda = \frac{1}{T_0}, \quad \boxed{E[\text{POISSON}] = \lambda} = \frac{1}{T_0}$$

DURATA DI UN SINGOLO ATTACCO

$$\boxed{T_0 = \frac{1}{\lambda}}$$

2)Dobbiamo calcolare il tempo in cui stiamo in ciascuna fase del ciclo.
Gli attacchi contro il nodo seguono una distribuzione geometrica, quindi:

$$P(X=n) = P(1-P)^{n-1} \qquad E[\text{GEOMETRICA}] = \frac{1}{P}$$

INFETTATO DOPO n ATTACCHI

$$P(X=n)\ \underline{(1-\alpha)} \cdot \underline{(\alpha)^{n-1}}$$

PROB. DI SUCCESSO ATTACCO

$$E[P(X=n)] = \frac{1}{1-\alpha}$$

NUMERO MEDIO DI ATTACCHI NECESSARI PER COMPROMETTERE IL NODO

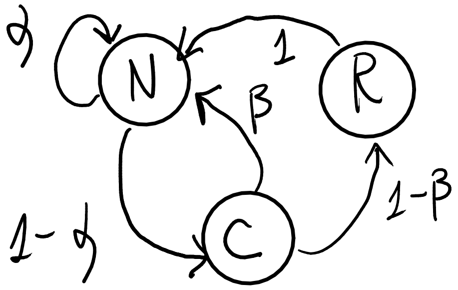
Quindi siccome sappiamo la durata media di un attacco e il numero medio di attacchi per compromettere il server possiamo calcolare il tempo medio in cui rimaniamo nello stato normale (non compromesso)

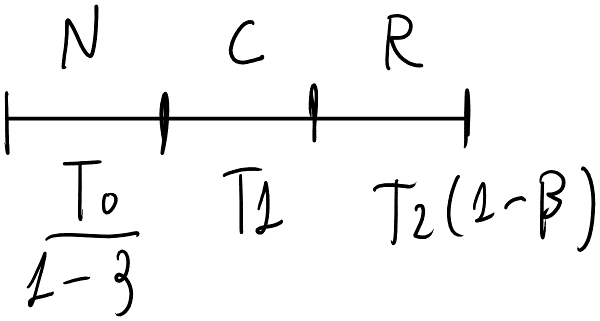
$$T_N = \frac{1}{1-\alpha} \cdot T_0$$

Per quanto riguarda T_r invece, il tempo in cui il nodo viene ripristinato manualmente, abbiamo che la quantità di tempo in cui ci stiamo è:

$$T_R = T_2(1-\beta) + \cancelto{0}{0 \cdot \beta}$$

3) Adesso che abbiamo calcolato tutti i tempi, possiamo scrivere il renewal cycle





Calcoliamo la fraction of time the node is not handling any traffic ossia P(R)

$$P[R] = \frac{E[R]}{E[\text{CICLO}]} = \frac{T_2(1-\beta)}{\frac{T_0}{1-\gamma} + T_1 + T_2(1-\beta)}$$

Calcoliamo average traffic per unit time

$$E[\text{OUTPUT}] = \frac{\sum \text{TEMPOSTATO} \cdot \text{TRAFFICO\_STATO}}{\text{TEMPO CICLO}} = \frac{\frac{T_0}{1-\gamma} \cdot 10 + T_1 \cdot 3}{\frac{T_0}{1-\gamma} + T_1 + T_2(1-\beta)}$$

(b) Compute how often (e.g., how many times a day on average) a human operator's intervention is needed.

1) Calcoliamo quanti cicli ci sono in un giorno

$$\#\text{ CICLI/GIORNO} = \frac{24 \cdot 60}{E[\text{CICLO}]} = \frac{24 \cdot 60}{\frac{T_0}{1-\gamma} + T_1 + T_2(1-\beta)}$$

2) Calcoliamo il numero di interventi al giorno

$$\text{FREQ\_INT} = \#\text{ CICLI} \cdot \text{PROB INTERVENTO IN UN CICLO} = \#\text{ CICLI/GIORNO} \cdot (1-\beta)$$

3)Calcoliamo ogni quanti giorni interviene l'operatore

$$T = \frac{1}{\text{FREQ}} \Rightarrow \frac{1}{\text{FREQ\_INT}} = 7{,}2 \text{ GIORNI}$$

RENEWAL CYCLE (2 SERVER)

E4 Consider a node that contains two identical and independent servers, each able to stream data at 1 Gbps. Each server is subject to attacks according to a Poisson process with rate $\lambda = 10$ attacks/hour, and each attack is effective with probability 1/9, whereas it has no consequences with probability 8/9 (Hint: only consider the process of *effective* attacks). As a result of each effective attack, the server will remain inoperational (i.e., with zero streaming rate) for an exponential time with average $T = 6$ minutes, during which any arriving attack will have no effect, and then will resume normal operations.

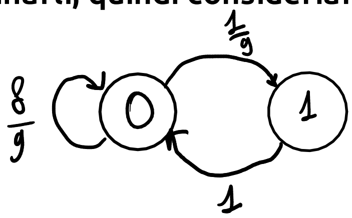
(a) Compute the fraction of time during which the node does not stream any data (i.e., both servers are inoperational), and the average duration of a period of time during which no data is streamed

(b) by considering an appropriate renewal cycle, compute the average duration of the time interval during which the node is able to stream data without interruptions (i.e., there is always at least one server working)

(c) compute the average total streaming rate of the node in Gbps.

(a) Compute the fraction of time during which the node does not stream any data (i.e., both servers are inoperational), and the average duration of a period of time during which no data is streamed

In realtà, quando abbiamo due processi indipendenti, ci merita analizzarli prima individualmente e poi combinarli, quindi consideriamo prima se fosse un solo server. Quindi calcoliamo i vari tempi del ciclo



$$\lambda = 10 \text{ ATTACCHI}/h \Rightarrow \frac{10}{60} \Rightarrow \frac{1}{6} \text{ ATTACCHI/MIN}$$

$$T_{\text{DURATA MEDIA ATTACCO}} = \frac{1}{\lambda} = 6 \text{ MINUTI}$$

Calcoliamo la il numero medio di attacchi prima che ci sia un successo

$$P(X=m) = P(1-P)^{m-1} \qquad E[\text{GEOMETRICA}] = \frac{1}{P}$$

INFETTATO DOPO m ATTACCHI

$$P(X=m) \; \underbrace{\left(\frac{1}{9}\right)} \cdot \left(\frac{8}{9}\right)^{m-1}$$

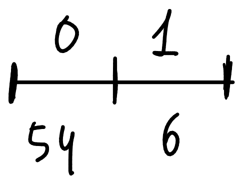
PROB DI SUCCESSO ATTACCO

$$\Rightarrow \boxed{E[\text{GEOMETRICA}] = \frac{1}{P} =} \; 9 \Rightarrow$$ NUMERO MEDIO ATTACCHI PER AVERE SUCCESSO

Calcoliamo il tempo di funzionamento normale

$$T_0 = \text{TEMPO MEDIO ATTACCO} \times \text{N MEDIO ATTACCHI} = 6 \cdot 9 = 54 \text{ MIN}$$

Il nostro ciclo per un singolo nodo è quindi:



Quindi:

$$P[1 \text{ SERVER DOWN}] = \frac{6}{54+6} = \frac{1}{10} \qquad P[2 \text{ SERVER DOWN}] = \left(\frac{1}{10}\right)^2 = \frac{1}{100}$$

Mentre l'expectation che entrambi non funzionino è:

$$\boxed{E[\text{ENTRAMBI DOWN}] = \frac{1}{2} \cdot T_2} = \frac{1}{2} \cdot 6 = 3 \text{ MIN}$$

(b) by considering an appropriate renewal cycle, compute the average duration of the time interval during which the node is able to stream data without interruptions (i.e., there is always at least one server working)

$$P[\text{ENTRAMBI DOWN}] = \frac{E[\text{ENTRAMBI DOWN}]}{E[\text{CICLO}]}$$

$$\Rightarrow E[\text{CICLO}] = \frac{E[\text{ENTRAMBI DOWN}]}{P[\text{ENTRAMBI DOWN}]} = \frac{3}{\frac{1}{100}} = 300 \text{ MIN}$$

$$E[\text{ALMENO 1 FUNZIONA}] = E[\text{NOT ENTRAMBI DOWN}]$$

$$= E[\text{CICLO}] \cdot (1 - P(\text{ENTRAMBI DOWN})) = 300 \cdot \left(1 - \frac{1}{100}\right) = 297 \text{ MIN}$$

(c) compute the average total streaming rate of the node in Gbps.

$$\text{STREAMING RATE} = (1+1) \cdot P[\text{ENTRAMBI UP}] + 2\left(1 \cdot P[1 \text{ UP}] \cdot P[1 \text{ DOWN}]\right)$$

$$= 2 \cdot \left(\frac{9}{10}\right)^2 + 2\left(\frac{9}{10}\right) \cdot \left(\frac{1}{10}\right) = 1,62 + 0,18 = 1,8$$

M/G/∞

E3 Consider a frequency division transmission system in which the number of channels is so large that the probability they are all occupied is negligible. Such system receives connection requests according to a Poisson process with rate $\lambda = 100$ calls per hour, and the duration of each call is exponential with mean 6 minutes. Let $X(t)$ be the number of occupied channels at time t.

(a) Compute the average of $X(t)$ at $t = 6, 10$ minutes and for $t = \infty$.

(b) Compute $P[X(t) = 10]$ for $t = 6$ and $t = \infty$

(c) Repeat the previous calculations assuming that the call duration is uniformly distributed in $[2, 10]$ (minutes)

G(t) → PROB OF LEAVING THE SERVICE

(a) Compute the average of $X(t)$ at $t = 6, 10$ minutes and for $t = \infty$.

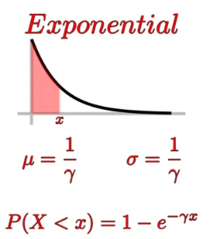
$$\lambda = 100 \text{ CALL}/h \Rightarrow \frac{10\cancel{0}}{6\cancel{0}} = \frac{5}{3} \text{ CALL/MIN}$$

Il numero di elementi presenti in una coda M/G/inf è descritto da una Poisson a media lambda_pt

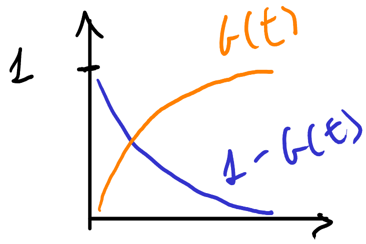
$$\boxed{\lambda_{PT} = \lambda \cdot \int_0^t [1 - G(y)]\, dy}$$

FUNZIONE DELLA DURATA PERMANENZA (PROB DI ESSERE NEL PROCESSO)

Dove però 1- G(y) è però calcolato facendo il not probabilistico della funzione G(y)



in questo caso la distribuzione è esponenziale, quindi:



$$\mu = 6 \text{ MIN} \Rightarrow \mu = \frac{1}{\gamma} \rightarrow P = \frac{1}{6}$$

$$G(x) = 1 - e^{-\frac{1}{6}x}$$

$$\Rightarrow 1 - G(x) = 1 - (1 - e^{-\frac{1}{6}x}) = e^{-\frac{1}{6}x}$$

Adesso possiamo calcolare lambda_pt

$$\lambda_{PT} = \frac{5}{3} \int_0^t e^{-\frac{1}{6}y}\, dy = -\frac{5}{3} \cdot 6 \cdot \left(e^{-\frac{1}{6}t} - 1\right) = 10\left(1 - e^{-\frac{t}{6}}\right)$$

Il testo chiede la media, ma noi sappiamo che in una poisson $E[\text{POISSON}] = \lambda$

quindi ci basta calcolare lambda_pt e abbiamo la media

$t = 6$: $\lambda_{PT} = 10 \cdot (1 - e^{-1}) = 6{,}32$

$t = 10$: $\lambda_{PT} = 10\,(1 - e^{-\frac{5}{3}}) = 8{,}11$

$t \to +\infty$: $\lambda_{PT} = \lim_{t \to \infty} 10\,(1 - \cancel{e^{-\frac{t}{6}}}^{\,0}) = 10$

(b) Compute $P[X(t) = 10]$ for $t = 6$ and $t = \infty$

In questo caso la distribuzione che descrive il numero di persone presenti al tempo è semplicemente una poisson con parametro lambda_pt

$$\boxed{P[X(t)=m] = e^{-\lambda_{PT}} \cdot \frac{\lambda_{PT}^{m}}{m!}}$$

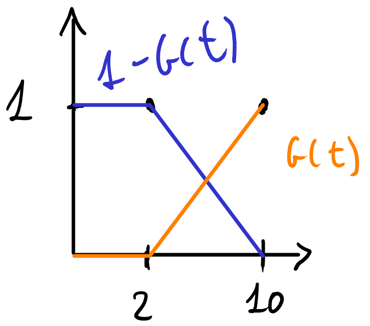
Prendiamo come lambda pt quelle relative ai tempi calcolate prima

$$P[X(6)=10] = e^{-6,33} \cdot \frac{(6,33)^{10}}{10!} = 0,05$$

$$P[X(\infty)=10] = e^{-10} \cdot \frac{(10)^{10}}{10!} = 0,125$$

(c) Repeat the previous calculations assuming that the call duration is uniformly distributed in [2, 10] (minutes)

L'unica cosa che cambia è la funzione della durata della permanenza, che diventa:



$$1-G(t) \Rightarrow P(2,1) \quad P(10,0)$$

$$y-0 = \frac{-1}{2-10}(x-10) \Rightarrow y = -\frac{1}{8}x + \frac{5}{4}$$

Unica cosa a cui fare attenzione è quando si fa l'integrale per calcolare lambda_pt, ricordarsi di includere la prima parte

$$\lambda_{PT} = \frac{5}{3}\left[\int_0^2 1\,dz + \int_2^t -\frac{1}{8}x + \frac{5}{4}\,dz\right]$$

poi è tutto uguale

E4 Consider an exhibition where visitors arrive according to a Poisson process with rate $\lambda = 12$ customers per hour. Each visitor spends a time uniformly distributed between 10 and 15 minutes, and then leaves. The room in which the exhibition is shown is large enough to ensure there is never a need to block customers at the entrance due to too many people inside. The exhibition is open from 8 AM to 6 PM.

(a) Compute the probability that fewer than two visitors arrive during the first fifteen minutes.

(b) Compute the probability that at 8:15 AM there is only one visitor in the room, and the probability that at closing time (6 PM) the room is empty.

$$\lambda = 12 \text{ customer}/h = \frac{12}{4} \text{ customer}/15\text{min}$$

(a) Compute the probability that fewer than two visitors arrive during the first fifteen minutes. $\rightarrow$ **arrive, quindi basta usare il lambda della poisson degli arrivi standard**

$$P[X(t)<2] = P[X(t)=0] + P[X(t)=1] = \frac{3^0}{0!} \cdot e^{-3} + \frac{3^1}{1!} \cdot e^{-3} = 4e^{-4}$$

(b) Compute the probability that at 8:15 AM there is only one visitor in the room, and the probability that at closing time (6 PM) the room is empty.

Calcoli lambda_pt al tempo 15 min e al tempo 600 min e poi applichi la solita formula. Il lambda pt è lo stesso, perché son tutti tempi maggiori di 15m, quindi quelli che una finestra completa

$$\lambda_{PT} = \lambda \int 1-G(z)\,dz = \frac{5}{2} \Rightarrow P(X(15)=1) = e^{-\frac{5}{2}} \cdot \frac{5}{2}$$

$$P(X(600)=0) = e^{-\frac{5}{2}}$$

Corso di Modelli e Analisi delle Prestazioni nelle Reti – AA 2006/2007
prova scritta – 05 settembre 2007 – parte A

E1 Consider a web server which receives download requests according to a Poisson process with rate $\lambda = 20$ requests per second. Each requests, after a fixed processing time of 20 ms, triggers the transfer of a file with size uniformly distributed between 1 and 2 MBytes. A request is said to be "active" from when it arrives to when the corresponding file transfer is completed. Assume that the server capacity, in terms of how many simultaneous requests it can handle, is infinite, and that the transfer data rate for each file is 100 Mbit/s, regardless of the number of files that are being transferred at any given time.

(a) Assuming that the server is switched on at time $t = 0$, when does the statistics of the number of active requests in the system reach its steady-state condition? In such condition, express $P[k$ active requests$]$

(b) Given that in an interval of duration T the system received N requests, find the probability that at the end of such interval there are no active requests in the two cases (b1) $T = 0.1$ s, $N = 2$ and (b2) $T = 1$ s, $N = 20$.

(a) Assuming that the server is switched on at time $t = 0$, when does the statistics of the number of active requests in the system reach its steady-state condition? In such condition, express $P[k$ active requests$]$

Calcoliamo l'intervallo di service time una volta entrati nel processo

$$\text{TRANSFER RATE} = 100\ \text{Mbps} = \frac{100}{8} = 12{,}5\ \text{MBps}$$

$$20 + \frac{U[1,2]}{TR} = [20+80,\ 20+160] = \overbrace{[100,\ 180]}^{\text{SERVICE TIME}}\ \text{ms}$$

$$E[\text{SERVICE TIME}] = \frac{100+180}{2} = 140\ \text{ms}$$

$$P[N(t)=k,\ t \geq 180\ \text{ms}] = \frac{\mu^k}{k!} e^{-\mu} \qquad \mu = \lambda \cdot E[\text{SERVICE TIME}]$$

$$\lambda = 20/s \rightarrow 0{,}02/ms \Rightarrow \mu = 0{,}02 \cdot 140 = 2{,}8$$

$$\Rightarrow P[N(t)=k,\ t \geq 180\ \text{ms}] = \frac{(2{,}8)^k}{k!} \cdot e^{-2{,}8}$$

Per la steady state condition, moltiplichiamo il tempo medio per 3, arbitrariamente

$$\boxed{T_{SS} = 3 \cdot E[\text{SERVICE TIME}]} \qquad T_{SS} = 3 \cdot 140 = 420\ \text{ms}$$

(b) Given that in an interval of duration T the system received N requests, find the probability that at the end of such interval there are no active requests in the two cases (b1) $T = 0.1$ s, $N = 2$ and (b2) $T = 1$ s, $N = 20$.

PROBABILITÀ CHE LA RICHIESTA SIA GESTITA ENTRO IL TEMPO T

$$\boxed{P[t_{ARRIVAL} + T_{PERMANENZA} \leq T] = \frac{1}{T} \cdot \int_0^T \overbrace{P[T_{PERMANENZA} \leq T - t]}^{1-G(t)}\, dt}$$

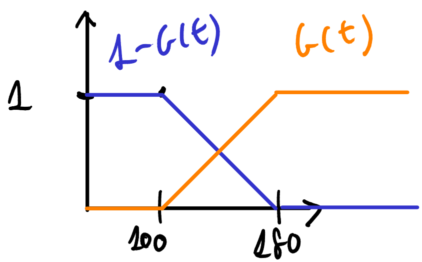
(b1) $T = 100$ ms

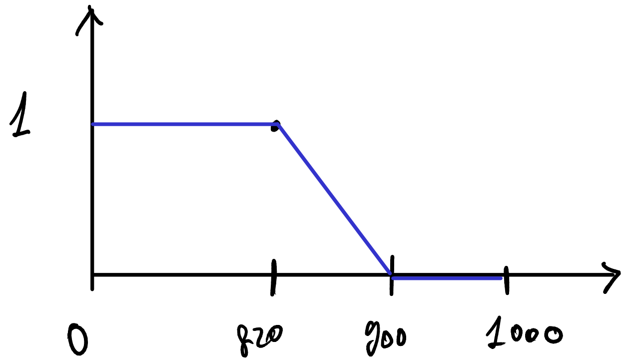
$P = 0$

ma il tempo minimo di gestione è 100ms, quindi è impossibile che al tempo 100ms la coda sia vuota se sono arrivati 2 processi durante l'intervallo

(b2) **Analizziamo il tempo di permanenza nel processo**

$A(100, 0)$ $B(180, 1)$





ARRIVALS $[0, 1000]$ ms

$$\frac{1}{1000}\left(\int_0^{820} 1\, dt + \int_{820}^{900} -\frac{x}{80} + \frac{45}{4}\, dt\right)$$

$$= \frac{1}{1000} \cdot \left(820 + \frac{80 \cdot 1}{2}\right) = \frac{860}{1000} = 0{,}86$$

PROBABILITÀ PROCESSO GESTITO ENTRO T

Vogliamo questa cosa per 20 processi, i processi sono indipendenti:

$$P[M(T) = 0 \mid X(T) = 20] = (0{,}86)^{20}$$

⁎ MARKOV CHAIN WITH TRANSIENT STATE

E3 Consider a Markov chain with the following transition matrix (states are numbered from 0 to 4):

$$P = \begin{array}{c|c} & \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0.4 & 0 & 0 & 0.6 \\ 0 & 0 & 0.5 & 0.2 & 0.3 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 & 0.7 \end{pmatrix} \end{array}$$

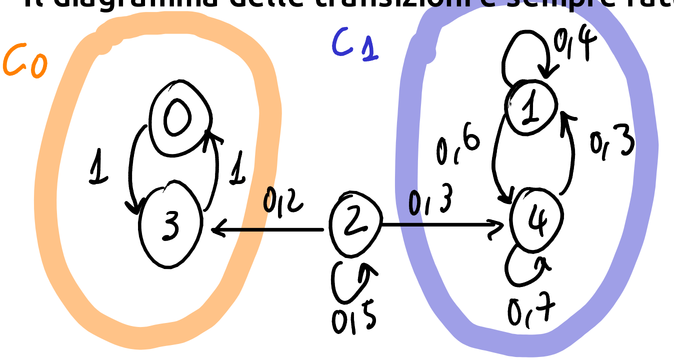
(a) Draw the transition diagram, identify the classes, classify the states, and compute the probabilities of absorption in all recurrent classes starting from each transient state

(b) compute $\lim_{n\to\infty} P^n$ and $\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} P^k$

(c) compute the average recurrence time for all states, and the average first passage time from any state to state 4.

(a) Draw the transition diagram, identify the classes, classify the states, and compute the probabilities of absorption in all recurrent classes starting from each transient state

Il diagramma delle transizioni è sempre fatto in questo modo:



$C_0 = \{0, 3\}$ **closed, recurrent, periodic with period d = 2**

$C_1 = \{1, 4\}$ **closed, recurrent, aperiodic**

$C_2 = \{2\}$ **transient**

Siccome una volta entrati in una closed class non ne usciamo più, a partire dall'unico transient state ci basta guardare le probabilità di entrare in ciascuna delle classi

$$P[\text{ABSORBED IN } C_0] = \frac{0{,}2}{0{,}2+0{,}3} = 0{,}4 \qquad P[\text{ABSORBED IN } C_1] = \frac{0{,}3}{0{,}2+0{,}3} = 0{,}6$$

(b) compute $\lim_{n\to\infty} P^n$ and $\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} P^k$

Dobbiamo valutare il comportamento del processo all'infinito. Valutiamo una classe chiusa per volta, per il momento possiamo calcolare soltanto i nodi che riguardano la classe aperiodica, in questo caso C2. Sulle colonne di C0 mettiamo delle x che calcoleremo dopo.

$$\begin{cases} \pi_1 = 0{,}4\pi_1 + 0{,}3\pi_4 \\ \pi_4 = 0{,}7\pi_4 + 0{,}6\pi_1 \\ \pi_1 + \pi_4 = 1 \end{cases} \quad \begin{cases} 0{,}6\pi_1 = 0{,}3\pi_1 - 0{,}3 \\ \\ / \end{cases} \quad \begin{cases} \pi_1 = 0{,}1 \\ \pi_4 = 0{,}9 \end{cases}$$

I valori dei pi_greco li mettiamo in modo simmetrico sulla matrice. Inoltre, sicuramente sappiamo che non saremo nello stato transiente, quindi di default la sua probabilità sarà zero.

$$\lim_{n\to\infty} P^n = \begin{array}{c|c} & \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{pmatrix} x & 0 & 0 & x & 0 \\ 0 & 0{,}1 & 0 & 0 & 0{,}9 \\ x & 0{,}06 & 0 & x & 0{,}54 \\ x & 0 & 0 & x & 0 \\ 0 & 0{,}1 & 0 & 0 & 0{,}9 \end{pmatrix} \end{array}$$

Possiamo anche calcolare le proabilità di andare dallo stato transiente nei nodi della classe aperiodica

$$P_{21}^{\infty} = P[\text{ANDARE IN } C_1] \cdot \pi_1 = 0{,}6 \cdot 0{,}1 = 0{,}06$$

$$P_{24}^{\infty} = P[\text{ANDARE IN } C_1] \cdot \pi_4 = 0{,}6 \cdot 0{,}9 = 0{,}54$$

Quella che abbiamo calcolato è la matrice per il lim di P^n all'infinito

Ora calcoliamo il limite della sommatoria includendo gli stati periodici:

$$\begin{cases} \pi_0 = 2 \cdot \pi_3 \\ \pi_3 = 1 \cdot \pi_0 \\ \pi_0 + \pi_3 = 1 \end{cases} \quad \begin{cases} \pi_0 = \frac{1}{2} \\ \pi_3 = \frac{1}{2} \end{cases} \Rightarrow \begin{array}{l} P^{\infty}_{20} = 0,4 \cdot \frac{1}{2} = 0,2 \\ P^{\infty}_{23} = 0,4 \cdot \frac{1}{2} = 0,2 \end{array}$$

$$\lim_{m \to \infty} \sum_{k=0}^{m-1} P^k = \begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 1 & 0 & 0,1 & 0 & 0 & 0,9 \\ 2 & 0,2 & 0,06 & 0 & 0,2 & 0,54 \\ 3 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 4 & 0 & 0,1 & 0 & 0 & 0,9 \end{array}$$

(c) compute the average recurrence time for all states, and the average first passage time from any state to state 4.

L'average recurrence time è definito come:

$$\boxed{R_i = \min\{ n > 0 : X_n = i \}} \quad \boxed{m_i = E[R_i]}$$

$$\boxed{m_i = \frac{1}{\pi_i}}$$

SOLO PER STATI POSITIVE RECURRENT APERIODIC CLOSED

$m_0 = 2$
$m_3 = 2$) GARANTITI

$m_1 = \frac{1}{\pi_1} = \frac{1}{0,1} = 10 \qquad m_4 = \frac{1}{0,9} = 1,11$

$m_2 = \infty$) LA MEDIA VA ALL'INFINITO PERCHÉ IN MOLTI CASI NON CI TORNEREMO MAI

Chiede l'average first passage time da qualsiasi stato allo stato 4

$\theta_{04} = \infty \quad \theta_{34} = \infty \quad \theta_{24} = \infty$) STESSO DISCORSO DI PRIMA C'È POSSIBILITÀ DI NON ARRIVARCI MAI

$\theta_{44} = \frac{1}{\pi_i} = \frac{1}{0,9}$) SOLO PER TORNARE IN SÉ STESSO

$\theta_{1,4} = 1 + \pi_{11} \cdot \theta_{1,4} = 1 + 0,1 \cdot \theta_{1,4} \Rightarrow \theta_{14} = \frac{1}{0,9}$

AVERAGE FIRST PASSAGE TIMES IF $i \neq j$ → $\boxed{\theta_{ij} = 1 + \sum_{k \neq j} P_{ik} \cdot \theta_{kj}}$

IF $i = j$ → $\boxed{\theta_{ii} = \frac{1}{\pi_i}}$

* MARKOV CHAIN WITH LONG RUN

Corso di Modelli e Analisi delle Prestazioni nelle Reti – AA 2006/2007
prova scritta – 09 luglio 2007 – parte A

E1 Consider a Markov chain X_n with the following transition matrix (states are numbered from 0 to 2), and initial state $X_0 = 0$:

$$P = \begin{pmatrix} 0.4 & 0.4 & 0.2 \\ 0.2 & 0.2 & 0.6 \\ 1 & 0 & 0 \end{pmatrix}$$

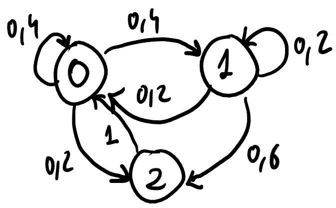
(a) Draw the transition diagram, and find the probability distributions of X_1, X_2 and X_{500}

(b) Compute the average first passage times from states 0, 1 and 2 to state 2.

(c) Compute $P[X_1 = 1, X_3 = 1 | X_2 = 1]$ and $P[X_2 = 1 | X_1 = 1, X_3 = 1]$.

(a) Draw the transition diagram, and find the probability distributions of X_1, X_2 and X_{500}

Lo stato di partenza è lo stato zero, quindi saranno per forza vettori riga



$$X_1 = \begin{pmatrix} 0,4 & 0,4 & 0,2 \end{pmatrix}$$

$$P^2 = \begin{pmatrix} 0.4 & 0.4 & 0.2 \\ 0.2 & 0.2 & 0.6 \\ 1 & 0 & 0 \end{pmatrix} \times \begin{pmatrix} 0.4 & 0.4 & 0.2 \\ 0.2 & 0.2 & 0.6 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0,44 & 0,24 & 0,32 \\ 0,72 & 0,12 & 0,16 \\ 0,4 & 0,4 & 0,2 \end{pmatrix}$$

$$X_2 = \begin{pmatrix} 0,44 & 0,24 & 0,32 \end{pmatrix}$$

$$\boxed{X_n = [\pi_0, \dots, \pi_K]}$$ FOR LARGE n

$$\begin{cases} \pi_0 = 0,4\,\pi_0 + 0,2\,\pi_1 + \pi_2 \\ \pi_1 = 0,2\,\pi_1 + 0,4\,\pi_0 \\ \pi_2 = 0,2\,\pi_0 + 0,6\,\pi_1 \\ \pi_1 + \pi_2 + \pi_3 = 1 \end{cases} \Rightarrow X_{500} = \begin{pmatrix} \pi_0 & \pi_1 & \pi_2 \end{pmatrix} = \begin{pmatrix} 0,5 & 0,25 & 0,25 \end{pmatrix}$$

(b) Compute the average first passage times from states 0, 1 and 2 to state 2.

AVERAGE FIRST PASSAGE TIMES → $\boxed{\theta_{ij} = 1 + \sum_{k \neq j} P_{ik} \cdot \theta_{kj}}$ IF $i \neq j$ ELSE → $\boxed{\theta_{ii} = \frac{1}{\pi_i}}$

$$\theta_{22} = \frac{1}{\pi_2} = \frac{1}{0,25} = 4$$

$$\theta_{02} = 1 + 0,4 \cdot \theta_{02} + 0,4\,\theta_{12}$$

$$\theta_{12} = 1 + 0,2\,\theta_{12} + 0,2 \cdot \theta_{02}$$

(c) Compute $P[X_1 = 1, X_3 = 1 | X_2 = 1]$ and $P[X_2 = 1 | X_1 = 1, X_3 = 1]$.

LAW OF CONDITIONAL PROBABILITY

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}$$

$$P[X_1=1, X_3=1 \mid X_2=1] = \frac{P[X_1=1, X_2=1, X_3=1 \mid X_0=0]}{P[X_2=1 \mid X_0=0]}$$

$$= \frac{P_{01}\cdot P_{11}\cdot P_{11}}{P^{(2)}_{01}} = \frac{0{,}4\cdot(0{,}2)^2}{0{,}24}$$

$$P[X_2=1 \mid X_1=1, X_3=1] = \frac{P[X_1=1, X_2=1, X_3=1 \mid X_0=0]}{P[X_1=1, X_3=1 \mid X_0=0]}$$

$$= \frac{\cancel{P_{01}}\cdot P_{11}\cdot P_{11}}{\cancel{P_{01}}\cdot P^{(2)}_{11}} = \frac{(P_{11})^2}{P^{(2)}_{11}}$$

E2 Consider a Markov chain X_n with the following transition matrix (states are numbered from 0 to 2):

$$P = \begin{pmatrix} 0.3 & 0.5 & 0.2 \\ 0.5 & 0.3 & 0.2 \\ 0 & 0 & 1 \end{pmatrix}$$

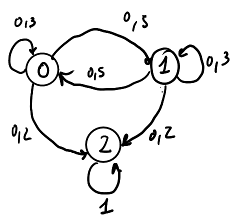
(a) Draw the transition diagram, and find the probability distribution of X_1, X_2 and X_{1000}, given $X_0 = 0$

(b) Let $W_{ij}^{(n)} = E\left[\sum_{k=0}^{n-1} I\{X_k = j\} \middle| X_0 = i\right]$ be the average number of visits to state j during the first n time slots, given that the chain starts in state i. Compute $\lim_{n\to\infty} W_{0j}^{(n)}$ for $j = 0, 1, 2$.

(c) Compute the average duration of the transient evolution of the chain, i.e., the time index at which the chain is absorbed.

(b) Let $W_{ij}^{(n)} = E\left[\sum_{k=0}^{n-1} I\{X_k = j\} \middle| X_0 = i\right]$ be the average number of visits to state j during the first n time slots, given that the chain starts in state i. Compute $\lim_{n\to\infty} W_{0j}^{(n)}$ for $j = 0, 1, 2$.

L'expected number of visits allo stato j partendo dallo stato i è definito come:



$$\boxed{V_{ij} = \delta_{ij} + \sum_{k \text{ TRANSIENT}} P_{ik}\cdot V_{kj}}$$

$$\delta_{ij} = \begin{cases} 1 & \text{IF } i=j \\ 0 & \text{OTHERWISE} \end{cases}$$

TRANSIENT $= \{0, 1\}$ REC. $= \{2\}$

Anche se chiede soltanto V0j dobbiamo fare il sistema e calcolarli tutti:

$$\begin{cases} V_{00} = 1 + P_{00}V_{00} + P_{01}V_{10} \\ V_{10} = 0 + P_{10}V_{00} + P_{11}V_{10} \\ V_{01} = 0 + P_{00}\cdot V_{01} + P_{01}\cdot V_{11} \\ V_{11} = 1 + P_{11}V_{11} + P_{10}\cdot V_{01} \end{cases}$$

risolvere a due a due e trovare i risultato

(c) Compute the average duration of the transient evolution of the chain, i.e., the time index at which the chain is absorbed.

L'expected number of steps until absorption a partire dallo stato i è definito come:

$$\boxed{T_i = \sum_{j \text{ TRANSIENT}} V_{ij}}$$

$$T_0 = V_{00} + V_{01}$$

$$T_1 = V_{10} + V_{11}$$

#TWO POISSON PROCESSES

E1 Consider two independent Poisson processes $X_1(t)$ e $X_2(t)$, where $X_i(t)$ is the number of arrivals for process i during $[0, t]$. The average number of arrivals per unit time of the two processes is $\lambda_1 = 1$ and $\lambda_2 = 1$, respectively.

(a) Compute $P[X_1(1) = 1 | X_1(2) + X_2(2) = 4]$ and $P[X_1(2) + X_2(2) = 4 | X_1(1) = 1]$

(b) Compute $P[X_1(1) + X_2(1) = 2 | X_1(2) = 0]$ and $P[X_1(1) + X_2(1) = 2 | X_1(2) = 1]$

Abbiamo informazioni su intervalli più grandi e vogliamo dedurre informazioni su intervalli più piccoli

- $0 < \mu < t$; $0 \leq k \leq m$; 2+ CONDIZIONI

(a) Compute $P[X_1(1) = 1 | X_1(2) + X_2(2) = 4]$

INFO SUL FUTURO

Theorem 4.5.6. *Let $X_1(t)$ and $X_2(t)$ be two independent Poisson processes with rates λ_1, λ_2 in the interval $(0, t)$. Let s be a subset of t s.t. $0 < s < t$. Given we know the total number n of arrivals in the interval $(0, t)$ i.e. $X_1(t) + X_2(t) = n$, the probability of $0 \leq k \leq n$ events occurring for process 1 in the subset $(0, s)$ is given by:*

$$\mathbb{P}[X_1(s) = k | X_1(t) + X_2(t) = n] = \frac{n!}{k!(n-k)!}\left(\frac{\lambda_1 s}{(\lambda_1 + \lambda_2)t}\right)^k \left(\frac{\lambda_1(t-s) + \lambda_2 t}{(\lambda_1 + \lambda_2)t}\right)^{n-k}$$

$0 < \mu < t \quad 0 \leq k \leq m$

$$P[X_1(\overset{\mu}{1}) = \overset{k}{1} | X_1(\overset{t}{2}) + X_2(\overset{t}{2}) = \overset{m}{4}]$$

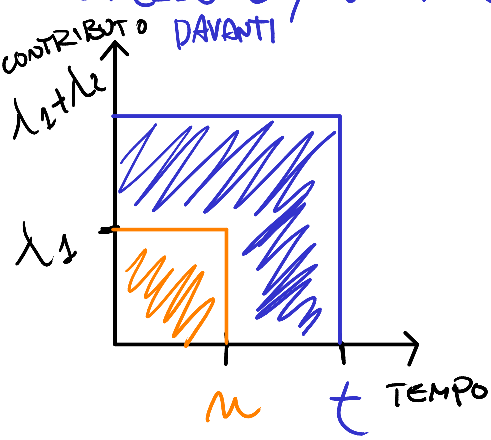
- $0 < \mu < t$; $0 \leq k \leq m$; 1 CONDIZIONE. $P[X_1(2) = 0 | X_1(3) = 1]$ INFO SUL FUTURO

$$P[X(\mu) = k | X(t) = m] = \left(\frac{m!}{k!(m-k)!}\right) \cdot \left(\frac{\mu}{t}\right)^k \cdot \left(1 - \frac{\mu}{t}\right)^{m-k}$$

- STESSO t ; $0 \leq k \leq m$; 2+ CONDIZIONI

$$P[X_1(t) = k | X_1(t) + X_2(t) = m] = \left(\frac{m!}{k!(m-k)!}\right) \cdot \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \cdot \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{m-k}$$

- STESSO t ; $0 \leq m \leq k$, $\mu < t$ INFO SUL PASSATO



$$P[X_1(t) + X_2(t) = k | X_1(\mu) = m]$$

$$\lambda_F = t \cdot (\lambda_1 + \lambda_2) - \mu \cdot \lambda_1$$

$$k_F = k - m$$

$$P[X_1(t) + X_2(t) = k - X_1(\mu) = m] = e^{-\lambda_F} \cdot \frac{\lambda_F^{k_F}}{k_F!}$$

- $P[X_1(2) + X_2(2) = 4 | X_1(1) = 1]$ $\quad P[X_1(1) + X_2(1) = 2 | X_1(1) = 1] = P[X_2(1) = 1]$

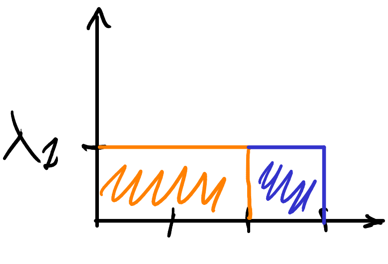
$\lambda_F = 2(\lambda_1 + \lambda_2) - 1 \cdot \lambda_1$

$k_F = 3 - 1$

- $P(X_1(3) = 3 | X_1(2) = 1)$

$\lambda_F = 3 \cdot \lambda_1 - 2\lambda_1 \quad \Rightarrow \quad e^{-\lambda_1} \cdot \frac{\lambda_1^2}{2!}$

$k_F = 3 - 1$



• STESSO t, $0 \le m \le k$, $t < \mu$ → INFO SUL FUTURO, VARI CASI

Vanno gestiti i vari casi, ossia valutare le probabilità del momento in cui sia arrivato il contributo

$$P[X_1(1) + X_2(1) = 2 \mid X_1(2) = 1]$$

$$= P[X_2(1) = 2] \cdot P[X_1(1) = 0 \mid X_1(2) = 1] + P[X_2(1) = 1] \cdot P[X_1(1) = 1 \mid X_1(2) = 1]$$

→ RISOLVI CON POISSON E $P[X(\mu)=k \mid X(t)=m]$

• STESSO t, $0 \le m \le k$, $t < \mu$ → CASO CONTRIBUTO ZERO

$$P[\cancel{X_1(1)} + X_2(1) = 2 \mid X_1(2) = 0] = P[X_2(1) = 2]$$